Crude oil, alkanes and alkenes

30 questions: 10 multiple choice, 20 structured.

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1 · Part 1

Paper 1 & 2 | Introduction | [1]

Which description fits the compounds in a homologous series?

A compounds with the same molecular formula but different structural formulae

B compounds that contain only carbon and hydrogen

C compounds with the same general formula and similar chemical properties

D compounds with the same boiling point but different functional groups

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2 · Part 1

Paper 1 & 2 | Introduction | [1]

The molecular formula of hexane is C_6H_{14}.

What is the empirical formula of hexane?

A C_3H_7

B CH_2

C C_6H_{14}

D C_2H_5

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3 · Part 1

Paper 1 & 2 Introduction [5]

Butane and 2-methylpropane are isomers. The structural formula of butane is $CH_3CH_2CH_2CH_3$.

(a) State what is meant by the term isomers. [2]

(b) Give the structural formula of 2-methylpropane. [1]

(c) State the molecular formula of 2-methylpropane. Give a reason for your answer. [2]

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4 · Part 1

Paper 1 & 2 Introduction [6]

Reactions of organic compounds can be classified as substitution, addition or combustion.

(a) The table shows two reactions of organic compounds. Complete the table by giving the type of each reaction.

Reaction	Equation	Type of reaction
1	$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$	
2	$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$	

[2]

(b) Reaction 3 is a substitution reaction: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$. Explain what is meant by a substitution reaction, using reaction 3 as an example. [2]

(c) Ethene, C_2H_4, reacts with hydrogen to form ethane, C_2H_6. State the type of this reaction. Give a reason for your answer. [2]

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5 · Part 1 Paper 1 & 2 Introduction [6]

Organic compounds are named using IUPAC rules.

(a) The table shows the structural formulae of four compounds. Complete the table by giving the IUPAC name of each compound.

Structural formula	IUPAC name
$CH_3CH_2CH(CH_3)CH_2CH_3$	
$CH_3CH(CH_3)CH(CH_3)CH_3$	
$CH_3CH_2CH{=}CHCH_3$	
$CH_3CH_2CH_2CH_2Cl$	

[4]

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(b) Priya named the compound $CH_3CH(CH_3)CH_2CH_2CH_3$ as 4-methylpentane. Explain why this name is not correct and give the correct IUPAC name. [2]

6 · Part 1

Paper 1 & 2 Introduction [6]

The table shows information about the first three members of the alkene homologous series.

Molecular formula	Boiling point in °C
C_2H_4	−104
C_3H_6	−47
C_4H_8	−6

(a) State what is meant by the term functional group. [1]

(b) Give two pieces of evidence from the table that these compounds are members of the same homologous series. [2]

(c) The next member of the series has five carbon atoms in each molecule. Predict its molecular formula and estimate its boiling point. [2]

(d) Explain why these three compounds have similar chemical reactions. [1]

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7 · Part 1

Paper 1 & 2 | Crude oil | [1]

Which fraction from crude oil is used to make surfaces for roads and roofs?

A fuel oil

B kerosene

C diesel

D bitumen

8 · Part 1

Paper 1 & 2 | Crude oil | [1]

Which statement about crude oil is correct?

A It is a single compound of carbon and hydrogen.

B It is a mixture of hydrocarbons.

C It is a mixture of the elements carbon and hydrogen.

D It is a mixture of alkenes only.

9 · Part 1

Paper 1 & 2 | Crude oil | [1]

Why is carbon monoxide poisonous?

A It reduces the capacity of the blood to transport oxygen.

B It reacts with water in the air to form an acid.

C It is a greenhouse gas that traps heat in the body.

D It reacts with oxygen in the lungs to form carbon dioxide.

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10 · Part 1

Paper 1 & 2 | Crude oil | [1]

Which conditions are used for the catalytic cracking of long-chain alkanes?

	Catalyst	Temperature in °C
A	iron	600–700
B	alumina	200–300
C	silica or alumina	600–700
D	phosphoric acid	300

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11 · Part 1

Paper 1 & 2 | Crude oil | [7]

Crude oil is separated into fractions by fractional distillation in a fractionating column.

(a) Describe how fractional distillation separates crude oil into fractions. [4]

(b) Explain why bitumen is collected at the bottom of the fractionating column. [2]

(c) Give a reason why the gasoline fraction does not have a single boiling point. [1]

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12 · Part 1

Paper 1 & 2 Crude oil [5]

Crude oil is separated into fractions that have different uses.

(a) The table shows some uses of fractions from crude oil. Complete the table by giving the name of the fraction for each use.

Use	Name of fraction
fuel for aircraft	
bottled gas for heating and cooking	
fuel for large ships and some power stations	

[3]

(b) State one use of the diesel fraction. [1]

(c) State one use of the gasoline fraction. [1]

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13 · Part 1

Paper 1 & 2 | Crude oil | [6]

Propane, C_3H_8, is used as a fuel in camping stoves.

(a) State what is meant by the term fuel. [1]

(b) Complete the balanced equation for the complete combustion of propane: C_3H_8 + ...O_2 → .....CO_2 + ...H_2O [2]

(c) When the air supply to the stove is limited, incomplete combustion takes place. Name two products that can form in incomplete combustion but do not form in complete combustion. [2]

(d) Suggest one observation that would show that incomplete combustion is taking place in the stove. [1]

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14 · Part 1

Paper 1 & 2 | Crude oil | [5]

Car engines release nitrogen monoxide, NO, and other oxides of nitrogen into the air.

(a) Complete the equation for the formation of nitrogen monoxide: $N_2 + O_2 \rightarrow$ .....NO [1]

(b) State where the nitrogen and the oxygen come from. Explain why they react inside a car engine but not in the air around us. [2]

(c) Describe how oxides of nitrogen contribute to acid rain. [2]

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15 · Part 1

Paper 1 & 2 | Crude oil | [7]

Diesel fuel is obtained from crude oil. When some diesel fuels burn, sulfur dioxide is produced.

(a) Explain why burning some diesel fuels produces sulfur dioxide. [2]

(b) Give the balanced equation for the reaction of sulfur with oxygen to form sulfur dioxide. [1]

(c) The table shows the mass of sulfur in 1000 kg of two diesel fuels, X and Y. Deduce which fuel causes less acid rain when it is burned. Give a reason for your answer.

Fuel	Mass of sulfur in 1000 kg of fuel in kg
X	2.0
Y	0.010

[2]

(d) Calculate the mass of sulfur dioxide, in kg, formed when all the sulfur in 1000 kg of fuel X burns. Use the Periodic Table. [2]

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16 · Part 1

Paper 1 & 2 Crude oil [6]

In a laboratory, liquid paraffin (a mixture of long-chain alkanes) is cracked.

Mineral wool soaked in liquid paraffin is placed at the closed end of a boiling tube. Pieces of aluminium oxide are placed in the middle of the tube. A delivery tube leads from the boiling tube to an upside-down test tube filled with water, standing in a trough of water. The aluminium oxide is heated strongly, and the flame is moved to the mineral wool from time to time so that the paraffin vaporises and passes over the hot aluminium oxide. The gas produced is collected in the test tube.

(a) Give a reason why the aluminium oxide is heated strongly. [1]

(b) The first test tube of gas collected is thrown away. Suggest why. [1]

(c) Explain why the delivery tube must be removed from the water before heating is stopped. [2]

(d) One reaction that takes place is $C_{10}H_{22} \rightarrow C_6H_{14} + C_4H_8$. Identify the product that is an alkene. Give a reason for your answer. [2]

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17 · Part 1

Paper 1 & 2 Crude oil [7]

The table shows the supply of some fractions from a typical crude oil and the demand for these fractions.

Fraction	Supply as a percentage of the crude oil	Demand as a percentage of the total
refinery gases	4	6
gasoline	10	25
kerosene	13	8
diesel	18	23
fuel oil and bitumen	55	38

(a) Identify the fraction with the largest difference between demand and supply, where demand is greater than supply. [1]

(b) Explain, using data from the table, why oil refineries crack some of the fractions from crude oil. [4]

(c) Fuel oil contains the alkane $C_{20}H_{42}$. When one molecule of $C_{20}H_{42}$ is cracked, the only products are one molecule of C_8H_{18} and some molecules of ethene, C_2H_4. Determine the number of ethene molecules formed. Show your working. [2]

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18 · Part 1

Paper 1 & 2 | Crude oil | [6]

Candle wax is burned to investigate the products of its combustion.

The lit candle is placed under an upside-down funnel. A pump draws the gases from the flame through a U-tube cooled in ice, and then through limewater.

(a) A colourless liquid collects in the U-tube. The liquid turns white anhydrous copper(II) sulfate blue. Identify the liquid. [1]

(b) State the change seen in the limewater and name the gas that causes it. [2]

(c) Explain what these results show about the elements in candle wax. [2]

(d) Give a reason why these results do not prove that candle wax is a hydrocarbon. [1]

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19 · Part 1

Paper 1 & 2 Crude oil [6]

The table shows information about some fractions from crude oil.

Fraction	Number of carbon atoms in each molecule	Boiling range in °C
refinery gases	1–4	below 30
gasoline	5–10	30–180
kerosene	10–16	180–260
diesel	14–20	260–340
fuel oil	20–40	above 340

(a) Describe the relationship between the number of carbon atoms in the molecules and the boiling range of the fraction. [1]

(b) Explain this relationship. [2]

(c) State how the colour and the viscosity of the fractions change from gasoline to fuel oil. [2]

(d) The alkane $C_{13}H_{28}$ is found in one of the fractions in the table. Identify this fraction. [1]

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20 · Part 1

Paper 1 & 2 Alkanes [1]

Each molecule of an alkane contains 9 carbon atoms.

What is the molecular formula of this alkane?

A C_9H_{18}

B C_9H_{16}

C C_9H_{22}

D C_9H_{20}

21 · Part 1

Paper 1 & 2 Alkanes [1]

Which compound is a saturated hydrocarbon?

A $CH_2=CH_2$

B $CH_3CH_2CH_3$

C CH_3CH_2Br

D $CH_2=CHCH_2CH_2CH_2CH_3$

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22 · Part 1 Paper 1 & 2 Alkanes [4]

There are three alkanes with the molecular formula C_5H_{12}. They are isomers.

(a) Name the isomer that has an unbranched chain of carbon atoms. [1]

(b) Give the structural formulae of the other two isomers. [2]

(c) Deduce the total number of covalent bonds in one molecule of any of these isomers. [1]

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23 · Part 1 Paper 1 & 2 Alkanes [5]

Propane, $CH_3CH_2CH_3$, reacts with bromine in the presence of ultraviolet radiation. The organic product has one bromine atom in each molecule. Hydrogen bromide, HBr, is also formed.

(a) Give the molecular formula of the organic product. [1]

(b) Two different isomers of this organic product can form. Give the structural formula of each isomer. [2]

(c) Give the IUPAC name of each isomer. [2]

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24 · Part 1 Paper 1 & 2 Alkanes [5]

Each molecule of an alkane contains 26 hydrogen atoms.

(a) Deduce the molecular formula of the alkane. [1]

(b) Calculate the relative formula mass (Mr) of the alkane. Use the Periodic Table. [1]

(c) Give the empirical formula of the alkane. [1]

(d) Explain why no alkane can have the empirical formula CH_2. [2]

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25 · Part 1 Paper 1 & 2 Alkenes [1]

Which molecular formula could be that of an alkene?

A C_6H_{14}

B C_2H_2

C C_7H_{14}

D C_8H_{18}

26 · Part 1 Paper 1 & 2 Alkenes [1]

Propene, $CH_2=CHCH_3$, reacts with bromine.

What is the structural formula of the product?

A $CH_2BrCHBrCH_3$

B $CH_3CH_2CH_2Br$

C $CH_2BrCH_2CH_2Br$

D $CH_3CBr_2CH_3$

27 · Part 1 Paper 1 & 2 Alkenes [5]

There are two alkenes with the molecular formula C_4H_8 that have an unbranched chain of carbon atoms.

(a) Give the structural formulae of these two alkenes. [2]

(b) Give the name of each of these two alkenes. [2]

(c) There is also an alkene with the molecular formula C_4H_8 that has a branched chain. Give its structural formula. [1]

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28 · Part 1 Paper 1 & 2 Alkenes [4]

Bromine water can be used to distinguish between hexane and hex-1-ene.

(a) Describe how you would carry out this test and give the result for each compound. [3]

(b) State the type of reaction that takes place between hex-1-ene and bromine. [1]

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29 · Part 1

Paper 1 & 2 Alkenes [6]

Ethene, C_2H_4, reacts with bromine at room temperature.

$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$

(a) Explain why ethene is described as an unsaturated hydrocarbon. [2]

(b) Give the IUPAC name of the product, $C_2H_4Br_2$. [1]

(c) In an experiment, 2.8 g of ethene reacts completely with bromine. Calculate the mass of bromine that reacts. Use the Periodic Table. [3]

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30 · Part 1

Paper 1 & 2 Alkenes [6]

J, K and L are hydrocarbons. J and L are alkenes. K is an alkane. The table shows the relative formula mass (Mr) of each hydrocarbon.

Hydrocarbon	Mr
J	84
K	100
L	70

Use the Periodic Table.

(a) Determine the molecular formula of J. Show your working. [2]

(b) Determine the molecular formula of K. Show your working. [2]

(c) L has an unbranched chain with the double bond between the first and second carbon atoms. Give the structural formula and the IUPAC name of L. [2]

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